Methaemoglobinaemia — Q49

Explain the 'saturation gap' and why pulse oximetry reads around 85%

- Saturation gap = difference between SaO2 calculated from PaO2 (normal) and measured SpO2/co-oximetry SaO2
- MetHb (Fe3+) cannot carry O2 and absorbs light equally at 660 and 940 nm → ratio ~1 → SpO2 reads ~85% regardless of true value
- PaO2 is normal because dissolved O2 is unaffected; co-oximetry needed for diagnosis

Describe the pathophysiology of methaemoglobinaemia and list common causes

- Haem iron oxidised from ferrous (Fe2+) to ferric (Fe3+) form
- Remaining haem shows increased O2 affinity (left shift) → impaired tissue O2 delivery
- Normally reduced by NADH–cytochrome b5 reductase (main) and NADPH–MetHb reductase (needs methylene blue)
- Causes: dapsone, local anaesthetics (benzocaine, prilocaine), nitrites/nitrates, nitric oxide, primaquine, rasburicase; congenital

Outline the treatment

- Stop the offending drug; high-flow O2
- Methylene blue 1–2 mg/kg IV over 5 min if symptomatic or MetHb > 20–30% (repeat after 30–60 min if needed)
- Contraindicated in G6PD deficiency (ineffective, causes haemolysis) → exchange transfusion or hyperbaric O2
- Dapsone has a long half-life (metabolites) → rebound; repeat doses; activated charcoal (multiple doses)
- Ascorbic acid as an alternative where methylene blue unavailable/contraindicated

Methaemoglobinaemia — Q50

Explain the mechanism of methylene blue in methaemoglobinaemia

- Methylene blue is reduced by NADPH–MetHb reductase to leukomethylene blue
- Leukomethylene blue reduces Fe3+ back to Fe2+ (non-enzymatically)
- NADPH comes from the pentose phosphate pathway (needs G6PD) – hence ineffective in G6PD deficiency

List the adverse effects and contraindications of methylene blue

- Serotonin syndrome with SSRIs/SNRIs/MAOIs (MB is a potent MAO-A inhibitor)
- Haemolysis in G6PD deficiency; paradoxical MetHb at high doses (> 7 mg/kg)
- Falsely low SpO2 (blue dye) for a short time; blue-green urine and skin
- Pulmonary hypertension worsening; hypertension; avoid in pregnancy (teratogenic)

When does methaemoglobinaemia fail to respond to methylene blue

- G6PD deficiency
- NADPH–MetHb reductase deficiency
- Sulfhaemoglobinaemia (irreversible; may be confused)
- Ongoing exposure/absorption (e.g., dapsone) or wrong diagnosis; consider exchange transfusion/HBO

Cyanide poisoning and hydroxocobalamin — Q51

Why do you suspect cyanide poisoning and how does cyanide cause toxicity

- Burning plastics, wool, silk release hydrogen cyanide; lactate > 8–10 mmol/L in a smoke victim strongly suggests cyanide
- Cyanide binds ferric iron in cytochrome c oxidase (complex IV) → blocks oxidative phosphorylation
- Cells switch to anaerobic metabolism → lactic acidosis; venous blood bright red (high SvO2, cells cannot extract O2)

Explain how hydroxocobalamin reverses cyanide toxicity and give the dose

- Hydroxocobalamin (vitamin B12a) has a cobalt ion that binds cyanide with high affinity
- Forms cyanocobalamin (vitamin B12), which is non-toxic and excreted in the urine
- Dose: 5 g IV over 15 min (70 mg/kg in children); repeat 5 g if needed
- Safe with CO poisoning (does not reduce O2 carrying), unlike nitrites

List the side effects of hydroxocobalamin and the rest of the management

- Red skin and urine; interference with colorimetric lab tests and haemodialysis machines (blood leak alarm)
- Hypertension (also scavenges NO); allergy
- 100% O2 / consider HBO for CO poisoning; early airway assessment for inhalational injury
- Supportive care, vasopressors, treat acidosis

Cyanide poisoning and hydroxocobalamin — Q52

Describe the alternative antidotes and their mechanisms

- Sodium nitrite (300 mg IV) / amyl nitrite → induces methaemoglobinaemia; MetHb binds cyanide (cyanomethaemoglobin) away from cytochrome oxidase
- Sodium thiosulfate (12.5 g IV) → sulfur donor for rhodanese, converting cyanide to thiocyanate (renally excreted)
- Dicobalt edetate → cobalt chelates cyanide; toxic if no cyanide present (oedema, arrhythmias)
- Thiosulfate is slow alone; best combined

Why are nitrites dangerous in smoke inhalation

- Smoke victims often have CO poisoning already reducing O2 carrying capacity
- Nitrites add MetHb → further loss of O2 carrying → worse tissue hypoxia
- Nitrites also cause vasodilatation and hypotension

Outline the supportive treatment

- Airway, 100% O2, ventilation
- Fluids and vasopressors for shock
- Benzodiazepines for seizures
- Correct acidosis; avoid mouth-to-mouth resuscitation; decontaminate if skin exposure

Methylene blue in vasoplegia — Q53

Define vasoplegia after cardiopulmonary bypass and its risk factors

- Low SVR shock with normal or high cardiac output despite adequate volume, after CPB
- Due to inflammatory response, excess NO, vasopressin deficiency, KATP channel activation
- Risk factors: long bypass time, preoperative ACEI/ARB, β-blockers, heparin, low EF, LVAD/transplant surgery

Which rescue therapy acts by inhibiting the nitric oxide/guanylate cyclase pathway? Explain its use

- Methylene blue
- Inhibits soluble guanylate cyclase (and NO synthase) → less cGMP → less vascular smooth muscle relaxation → raises SVR and MAP
- Dose 1–2 mg/kg IV over 20–60 min, then infusion if needed
- Avoid with serotonergic drugs, G6PD deficiency, pulmonary hypertension; interferes with SpO2 readings

List the other options for refractory vasoplegia

- Hydroxocobalamin 5 g IV (scavenges NO, inhibits NOS)
- Angiotensin II (ATHOS-3)
- Hydrocortisone
- Vasopressin if not already started; correct acidosis and hypocalcaemia

Toxic alcohol poisoning — Q54

Calculate the anion gap and osmolar gap and interpret them

- AG = 140 − (100 + 6) = 34 mmol/L → high anion gap metabolic acidosis
- Calculated osmolality = 2 × 140 + 5 + 5 = 290; osmolar gap = 330 − 290 = 40 mOsm/kg (normal < 10)
- High AG + high osmolar gap + low lactate → toxic alcohol
- Calcium oxalate crystals → ethylene glycol

Explain the metabolism of ethylene glycol and why it is toxic

- Ethylene glycol → (alcohol dehydrogenase) glycoaldehyde → glycolic acid (causes acidosis) → glyoxylic → oxalic acid
- Oxalate binds calcium → calcium oxalate crystals in kidneys (AKI), hypocalcaemia
- Parent compound causes osmolar gap; metabolites cause anion gap (gap shifts over time)

Outline the treatment, including the antidote

- Antidote: fomepizole (15 mg/kg then 10 mg/kg 12-hourly) or ethanol – blocks alcohol dehydrogenase
- Haemodialysis: severe acidosis (pH < 7.25), AKI, high levels, visual signs (methanol), deteriorating
- Bicarbonate to correct acidosis; calcium only if symptomatic hypocalcaemia
- Cofactors: thiamine and pyridoxine (ethylene glycol)

Toxic alcohol poisoning — Q55

Compare methanol and ethylene glycol poisoning

- Methanol → formaldehyde → formic acid; ethylene glycol → glycolic and oxalic acids
- Methanol: visual disturbance, optic disc oedema, blindness, basal ganglia (putamen) necrosis
- Ethylene glycol: CNS depression, cardiopulmonary failure, AKI with oxalate crystals, hypocalcaemia
- Both: early osmolar gap, later high AG acidosis

Explain why the osmolar gap may be normal in a late presentation

- Parent alcohol (osmotically active) has been metabolised
- Toxic acid metabolites raise the anion gap but not the osmolar gap much
- So a normal osmolar gap does not exclude toxic alcohol poisoning

List the indications for haemodialysis and cofactors used in methanol poisoning

- Severe acidosis, visual impairment, coma or seizures, AKI
- High methanol level (> 50 mg/dL; EXTRIP > 700 mg/L without fomepizole)
- Fomepizole/ethanol continued during dialysis (dose adjusted)
- Folinic acid/folic acid 50 mg IV 4–6 hourly (helps metabolise formate)

Local anaesthetic systemic toxicity — Q56

Describe the clinical features of LAST

- CNS: perioral numbness, metallic taste, tinnitus, agitation → seizures → coma
- CVS: hypertension/tachycardia early → bradycardia, wide QRS, VT/VF, asystole
- Bupivacaine is most cardiotoxic; cardiac arrest may occur without CNS warning

Calculate her dose and give the maximum safe dose of bupivacaine

- 40 mL × 5 mg/mL = 200 mg
- Max bupivacaine ~2 mg/kg (2.5 mg/kg with adrenaline) – 200 mg is above the limit for a 70 kg patient without adrenaline (~150 mg)

Outline the management of the cardiac arrest, including lipid emulsion dosing

- Stop injection, call for help, airway, 100% O2; control seizures with benzodiazepines
- 20% lipid emulsion 1.5 mL/kg IV over 1 min, then infusion 15 mL/kg/h
- Repeat bolus up to twice at 5-min intervals; double infusion to 30 mL/kg/h; max 12 mL/kg total
- Modified ALS: small adrenaline doses (≤ 1 µg/kg); avoid vasopressin, calcium channel blockers, β-blockers and local anaesthetics (lidocaine)
- Prolonged CPR (may need > 1 h); consider ECMO

Local anaesthetic systemic toxicity — Q57

Explain the proposed mechanisms of action of lipid emulsion

- Lipid sink: forms a lipid compartment in plasma that absorbs lipophilic drug away from tissues
- Lipid shuttle: carries drug to liver and muscle for storage/metabolism
- Direct cardiotonic effect: provides fatty acids to myocardium, improves calcium handling

List the factors that increase the risk of LAST

- High dose, intravascular injection, highly vascular sites (intercostal > epidural > brachial plexus)
- Extremes of age, low muscle mass, pregnancy
- Cardiac disease, conduction block, low EF
- Hepatic/renal failure, acidosis, hypoxia (more free drug)

Describe preventive measures

- Calculate and use the lowest effective dose; know maximum doses
- Ultrasound guidance; incremental injection with frequent aspiration
- Adrenaline test dose / adrenaline-containing solution as marker
- Monitoring during and 30 min after; lipid emulsion immediately available

Magnesium sulfate toxicity — Q58

Describe the clinical features of magnesium toxicity at different levels

- Therapeutic range ~2–4 mmol/L
- > 4–5 mmol/L: loss of deep tendon reflexes, nausea, flushing, somnolence
- > 6–7 mmol/L: respiratory depression, muscle paralysis
- > 7.5–12 mmol/L: heart block, cardiac arrest

What is the first treatment? Give the dose

- Stop the magnesium infusion
- Calcium gluconate 10% 10 mL (1 g) IV over 10 min – antagonises Mg at the neuromuscular junction and myocardium
- Support ventilation (bag–mask/intubation) if respiratory depression

Why did toxicity develop and how should it be monitored

- Mg is excreted renally – oliguria/AKI in pre-eclampsia leads to accumulation
- Monitor hourly: reflexes, RR (> 12), SpO2, urine output (> 30 mL/h)
- Serum Mg levels if renal impairment or clinical concern
- Reduce or stop infusion if oliguric; furosemide/dialysis to enhance clearance in severe cases

Magnesium sulfate toxicity — Q59

List the indications for IV magnesium in the ICU, with doses

- Eclampsia prevention/treatment: 4 g IV over 5–15 min, then 1 g/h for 24 h
- Torsades de pointes: 2 g IV over 10 min
- Acute severe asthma: 1.2–2 g IV over 20 min
- Hypomagnesaemia, AF rate control adjunct, refractory hypokalaemia

Describe the physiological actions of magnesium

- Calcium antagonist: blocks L-type calcium channels → smooth muscle relaxation, vasodilatation, bronchodilatation
- NMDA receptor antagonist (CNS), anticonvulsant in eclampsia
- Cofactor for Na/K-ATPase – needed to correct hypokalaemia
- Reduces acetylcholine release at NMJ → potentiates neuromuscular blockers

What are the causes and consequences of hypomagnesaemia in the ICU

- Causes: diuretics, PPIs, alcohol, diarrhoea, refeeding, aminoglycosides/amphotericin, CRRT
- Consequences: arrhythmias (torsades), refractory hypokalaemia and hypocalcaemia
- Weakness, seizures, tetany
- Associated with worse outcome and longer ventilation

Methylene blue in vasoplegia — Q60

Define refractory septic shock and outline your stepwise approach

- Persistent hypotension despite high-dose vasopressors (e.g., noradrenaline > 0.5 µg/kg/min) and adequate fluid
- Check source control, adequacy of antibiotics, fluid status, cardiac function (echo), acidosis and calcium
- Hydrocortisone 200 mg/day
- Add adrenaline (especially if cardiac dysfunction) or angiotensin II

Explain the mechanisms of vasodilatory shock

- Excess inducible NO synthase → NO → cGMP → vasodilatation
- Activation of KATP channels → hyperpolarisation of vascular smooth muscle
- Relative vasopressin deficiency
- Adrenergic receptor down-regulation and relative adrenal insufficiency

What is the evidence for methylene blue and angiotensin II in septic shock

- Methylene blue: small RCTs show faster shock resolution and less vasopressor need; no definitive mortality benefit
- Angiotensin II (ATHOS-3, 2017): raised MAP in vasodilatory shock; no mortality difference; risk of thrombosis (VTE prophylaxis)
- Both are rescue therapies in refractory shock